Where Our Post-Quantum VPN Is Used in the Real World

Final Year Project: Post-Quantum VPN Framework for Constrained Devices

November 2026

Summary

Our VPN lets a small, cheap device (an ESP32) send its data to a gateway through a locked tunnel. The lock is built to stay safe even when quantum computers arrive.

The best real-world use for it is **smart electricity meters**. A smart meter system has the same shape as our project: many small devices send data to one collector. Meters do not only encrypt. The stronger ones do the same three steps as our VPN: prove identity, agree on a key, then encrypt. The meter industry is now adding the same two algorithms that we used, ML-KEM and ML-DSA. We found no proof that such meters are installed in homes yet. So our project is early, but it is going where the industry is going.

Every argument in this report is followed by its source, with the page and paragraph where it can be found.

1 What our VPN does

Many small devices each open a tunnel to one gateway. Our VPN works in three steps:

1. **Prove identity.** The device and the gateway each prove who they are with an ML-DSA-44 signature.
2. **Agree on a key.** They agree on a fresh secret key with ML-KEM-512.
3. **Encrypt.** Every message is encrypted with AES-256-GCM.

This layout, with many small devices sending to one gateway, is a normal one for IoT systems. An earlier capstone thesis on a lightweight IoT VPN describes it like this: "a gateway-based IoT deployment in which hundreds to thousands of resource-constrained sensor nodes (e.g., ESP32-class devices) periodically transmitted telemetry over a wireless medium to a local sink/gateway."

Source: Mujtaba, Khowaja and Rizvi, *Lightweight Cryptography Based VPN Framework for Secure IoT Communication*, capstone thesis, Habib University, 2026, page 12, Section 2.2, paragraph 1, lines 1–3.

The same thesis says where this layout is used: "large-scale sensor networks used in smart-city and industrial monitoring deployments".

Source: same thesis, page 12, Section 2.2, paragraph 1, lines 9–10.

The devices are small. The thesis lists the ESP32 with 520 KB of memory, the largest of the six boards it compares.

Source: same thesis, page 15, Table 2.1, last row.

2 The application: smart electricity meters

A smart meter measures how much electricity a home uses and sends the readings to the electricity company. It does not send them directly. It sends them to a collector box nearby, called a *data concentrator*. The collector then sends them to the company's central system, called the *head-end system*.

Source: Seqred, "DLMS/COSEM protocol", section "AMI architecture in Poland", the diagram at the start of the section. https://seqred-website.pages.dev/en/blog/protokol-dlms-cosem/

Figure 1 shows our project and a smart meter system side by side. The shape is the same.

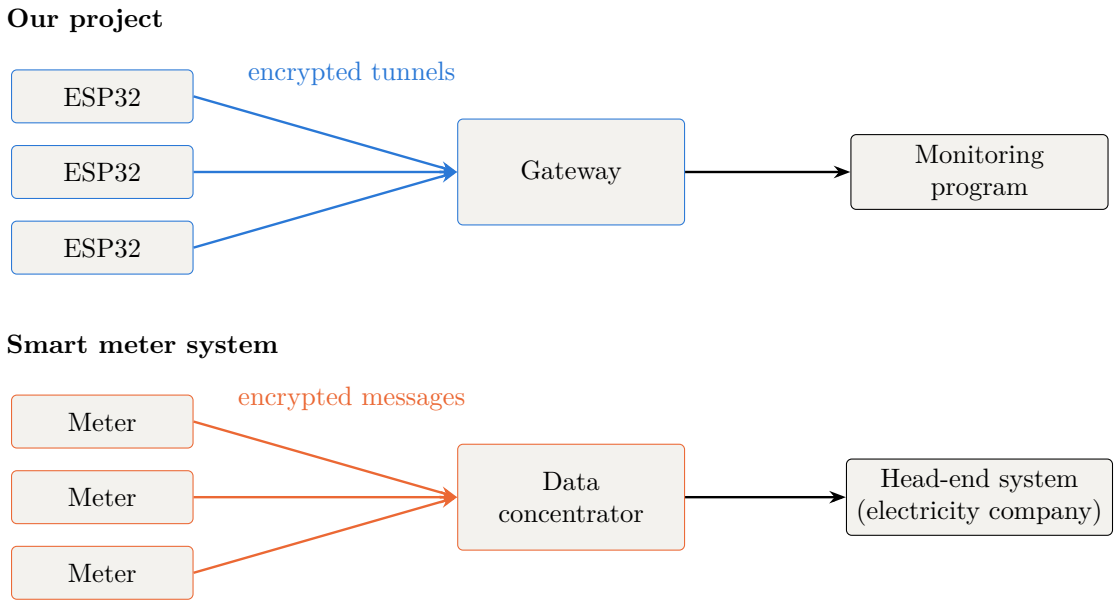


Figure 1: Our project and a smart meter system have the same shape: many small devices send protected data to one collector.

3 How smart meters are protected today

Meters use a standard called DLMS/COSEM. The protection is done by the meter's own software: it locks each message itself. Our VPN works the same way. The standard is described as covering "the device object model and the application layer".

Source: Seqred, "DLMS/COSEM protocol", opening section, the paragraph that begins "DLMS/COSEM is a two-part standard". https://seqred-website.pages.dev/en/blog/protokol-dlms-cosem/

The standard has three security levels, called Suite 0, Suite 1 and Suite 2.

Source: Huang, Chien, Wang and Liao, "The Implementation of an HSM-Based Smart Meter for Supporting DLMS/COSEM Security Suite 1", IoTBDS 2023, page 123, Section 1, right column, last paragraph. https://www.scitepress.org/Papers/2023/118253/118253.pdf

Most meters use the basic level: "Most of the smart meters deployed so far support the security mechanisms in Security Suite 0, including Advanced Encryption Standard-Galois/Counter Mode-128 (AES-GCM-128) for authenticated encryption and AES-128 key wrap for key transport." AES-GCM is the same kind of encryption our tunnel uses.

Source: Huang et al., IoTBDS 2023, page 124, left column, paragraph 1, lines 1–8. https://www.scitepress.org/Papers/2023/118253/118253.pdf

Stronger meters use Suite 1, which adds elliptic-curve algorithms: "ECDSA (Elliptic Curve Digital Signature Algorithm), ECDH (Elliptic Curve Diffie–Hellman key exchange), and SHA (Secure Hash Algorithm); these are defined in Security Suite 1."

Source: Huang et al., IoTBDS 2023, page 124, left column, paragraph 1, lines 15–18. https://www.scitepress.org/Papers/2023/118253/118253.pdf

4 Only encryption, or the full process like ours?

Smart meters do **not** only encrypt. A Suite 1 meter does the same three steps as our VPN, in the same order (Figure 2). It is not called a VPN, but it does the same job.

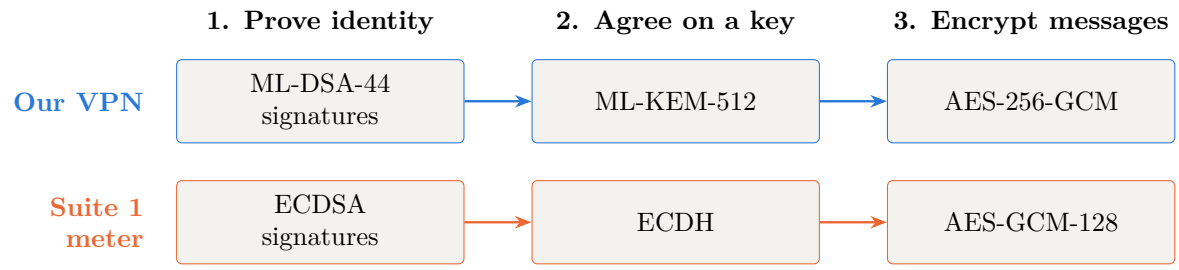


Figure 2: A Suite 1 smart meter and our VPN do the same three steps. Only the algorithms differ: ours are safe against quantum computers.

Step 1: prove identity. A meter must be checked before it can be used: "Establishing an AA is the first step in accessing smart meter services" and "Authentication takes place during AA establishment." (AA means *application association*, the connection between the two sides.)

Source: Huang et al., IoTBDS 2023, page 126, Section 4.1, paragraph 1. https://www.scitepress.org/Papers/2023/118253/118253.pdf

Both sides prove themselves with signatures, as in our VPN: "In Pass 3, the client needs to generate its authentication message and sign the message with its private key using ECDSA. In Pass 4, the server first verifies the received message from the client and then generate its authentication message using ECDSA."

Source: Huang et al., IoTBDS 2023, page 128, Section 4.4, the paragraph that begins "In Pass 3". https://www.scitepress.org/Papers/2023/118253/118253.pdf

Step 2: agree on a key. "ECDH is used for key agreement between client and server".

Source: Huang et al., IoTBDS 2023, page 127, Section 4.3, paragraph 1. https://www.scitepress.org/Papers/2023/118253/118253.pdf

Step 3: encrypt. The agreed key is then used to encrypt the messages: "AES-GCM-128 is the encryption algorithm for the general-ciphering APDU. The client and the server need to agree on a shared key for AES-GCM to encrypt and decrypt the APDU. This is achieved by using the ECDH algorithm". (An APDU is one message.)

Source: Huang et al., IoTBDS 2023, page 128, Section 4.5, paragraph 1. https://www.scitepress.org/Papers/2023/118253/118253.pdf

Most installed meters are simpler. They use Suite 0. They still check identity and encrypt, but they do not agree on a fresh key. The keys are stored in the meter and are sent to it in wrapped (locked) form. The paper says that in Suite 0 "authentication can be done using GMAC", and that keys are moved with "AES-128 key wrap for key transport".

Source: Huang et al., IoTBDS 2023, page 124, left column, paragraph 1, lines 6–11. https://www.scitepress.org/Papers/2023/118253/118253.pdf

The stored keys are listed as an "Authentication Key (AK) - key for HLS and authentication" and a "Master Key (MK) - master key, used for transporting other keys".

Source: Seqred, "DLMS/COSEM protocol", section "DLMS/COSEM security", the list of keys. https://seqred-website.pages.dev/en/blog/protokol-dlms-cosem/

Table 1 puts the levels side by side. The planned Suite 3 (Section 6) replaces ECDSA and ECDH with ML-DSA and ML-KEM, the two algorithms in our VPN.

Step	**Suite 0 (most meters)**	**Suite 1**	**Suite 3 (planned)**	**Our VPN**
1. Prove identity	GMAC with a stored key	ECDSA signatures, both sides	ML-DSA	ML-DSA-44 signatures, both sides
2. Agree on a key	No. Stored keys, sent wrapped	ECDH	ML-KEM	ML-KEM-512
3. Encrypt	AES-GCM-128	AES-GCM-128	not stated in our sources	AES-256-GCM

Table 1: The three steps in each smart meter security level and in our VPN. Sources: Suite 0 and Suite 1 from Huang et al. (pages 124 to 128, as cited above); Suite 3 from the SEALSQ article cited in Section 6.

5 Why this protection must change

Elliptic-curve algorithms can be broken by a quantum computer. ECDSA and ECDH, used in Suite 1, are of this kind. The US standards body NIST plans to stop their use. Its draft report marks the weaker versions as "Deprecated after 2030" and all versions as "Disallowed after 2035". This is a draft, so the dates could still change.

Source: NIST IR 8547 (initial public draft, November 2024), page 13, Table 2 (signatures, ECDSA row) and page 14, Table 4 (key establishment, Elliptic Curve row). https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf

Data recorded today can be opened later. NIST explains: "Encrypted data remains at risk because of the 'harvest now, decrypt later' threat in which adversaries collect encrypted data now with the goal of decrypting it once quantum technology matures."

Source: NIST IR 8547, page 1, Section 1, lines 105–107. https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf

A meter maker states the danger. Landis+Gyr, a large maker of smart meters, writes: "If encryption keys on smart meters are cracked, attackers could manipulate billing data or even disrupt control systems."

Source: Landis+Gyr, "The Quantum Risk: Why Utilities Must Act Now to Secure the Grid", 20 October 2025, section "Why Quantum Computing Threatens Grid Security", first bullet ("Meter Key Compromise"). https://www.landisgyr.com/us/en/home/knowledge/blog/2025/The-Quantum-Risk--Why-Utilities-Must-Act-Now-to-Secure-the-Grid

Meters stay in service for a long time. "Devices being designed today will remain deployed well into the 2040s." So a meter designed today will still be working after the old algorithms are no longer allowed (Figure 3).

Source: SEALSQ, "Why Smart Metering Security Must Evolve Beyond PQC", section "Requirements for Next-Generation Smart Meters", paragraph 1. https://www.sealsq.com/sealsq-blog/why-smart-metering-security-must-evolve-beyond-pqc

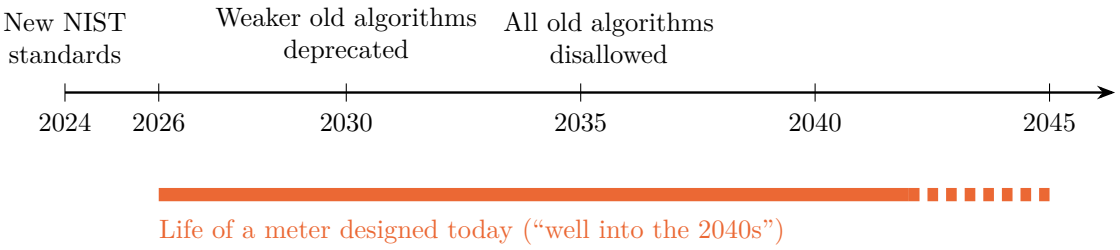


Figure 3: A meter designed today is still in service long after the old algorithms stop being allowed. Dates from the NIST draft and the SEALSQ article cited above; the 2024 standards are named in the Landis+Gyr article.

6 What the industry is doing

The meter standard is adding ML-KEM and ML-DSA. These are the two post-quantum algorithms we used. "With the DLMS User Association finalizing Suite 3, introducing ML-KEM for key encapsulation and ML-DSA for digital signatures". This comes from SEALSQ, a company that sells chips for it, and it points to the association's own 2025 position paper, which we could not open.

Source: SEALSQ, "Why Smart Metering Security Must Evolve Beyond PQC", opening paragraph. The position paper is named in section "Why Hardware Boundaries Matter", last paragraph. https://www.sealsq.com/sealsq-blog/why-smart-metering-security-must-evolve-beyond-pqc

Landis+Gyr names the same two algorithms. Its article lists "FIPS 203: ML-KEM (CRYSTALS-Kyber) for encryption" and "FIPS 204: ML-DSA (CRYSTALS-Dilithium) for digital signatures".

Source: Landis+Gyr, "The Quantum Risk", 20 October 2025, section "The Timeline Is Closer Than You Think", bullets 1 and 2. https://www.landisgyr.com/us/en/home/knowledge/blog/2025/The-Quantum-Risk--Why-Utilities-Must-Act-Now-to-Secure-the-Grid

It also says: "Software updates alone won't solve the quantum challenge", and points to a SEALSQ chip that has these algorithms built in.

Source: Landis+Gyr, "The Quantum Risk", section "Moving Forward: What Utilities Must Do Now", bullet 3 ("Adopt Hardware-Based Security"). https://www.landisgyr.com/us/en/home/knowledge/blog/2025/The-Quantum-Risk--Why-Utilities-Must-Act-Now-to-Secure-the-Grid

The work between these companies is large. SEALSQ reports that it "extended its relationship with Landis & Gyr to provide enhanced PKI services for AMI network and meter security, including a program to supply approximately 30 million digital certificates for Landis & Gyr smart meters in Asia-Pacific". (A certificate is a digital identity card for a meter.)

Source: SEALSQ Corp., *Half Year Report 2026*, page 30, paragraph 1, lines 1–5. https://www.sealsq.com/hubfs/SEALSQ%20H1%20Report%202026_version%2011.09.2026_Final3.pdf

Another company sells this for the small chips inside meters. PQShield offers ML-KEM and ML-DSA software for meters and explains why it is hard: "These devices, often running on ARM Cortex CPUs can be difficult to update due to their long lifecycles and low-power, resource constrained microcontrollers."

Source: PQShield, "ML-KEM and ML-DSA for ARM Cortex M CPU in smart meters", section "Upgrading smart meters with quantum security", paragraph 1. The software is described in section "PQMicroLibCore", paragraph 1. https://pqshield.com/?p=9909

7 VPNs that already exist

VPNs for small devices and post-quantum VPNs both exist already, but separately.

A classical VPN that runs on the ESP32. Husarnet is a VPN that works "even on ESP32 microcontrollers".

Source: Husarnet source code, README, feature list, bullet "Lightweight". https://github.com/husarnet/husarnet

Its algorithms are classical: "Husarnet uses X25519 from libsodium for key exchange" and "The packets are encrypted using libsodium ChaCha20-Poly1305". X25519 is not safe against a quantum computer.

Source: Husarnet, "How Husarnet works", section "Husarnet security", the paragraph that begins "Cryptography:". https://husarnet.com/docs/how-husarnet-works/

A post-quantum VPN on large routers. "Cisco and Fortinet are the first third-party vendors validated to interoperate with Cloudflare IPsec using ML-KEM". It "combines classical Diffie-Hellman (Group 20) with ML-KEM-768 or ML-KEM-1024".

Source: Cloudflare changelog, "Post-quantum IPsec interoperability with third-party devices", 30 April 2026, paragraphs 1 and 2. https://developers.cloudflare.com/changelog/post/2026-04-30-ipsec-post-quantum-third-party/index.md

Our project joins the two: a post-quantum VPN that runs on the small device itself.

8 How our project fits

Table 2 compares the planned smart meter security (Suite 3) with our VPN. The sources for the left column are given in Sections 2, 3, 4 and 6.

	Smart meters (planned Suite 3)	**Our VPN**
Layout	Many meters to one collector	Many ESP32s to one gateway
Where the locking happens	In the meter's own software	In the ESP32's own software
Proving identity	ML-DSA	ML-DSA-44
Agreeing on a key	ML-KEM	ML-KEM-512
Encrypting messages	AES-GCM in today's suites	AES-256-GCM
Hardware	Small ARM chip, plus a secure chip	ESP32 only

Table 2: The planned smart meter security and our VPN use the same building blocks.

What our project adds. We built the VPN in two versions, classical and post-quantum, and measured both on a real ESP32. The post-quantum version costs more only once, when the device connects:

- the handshake takes 219 ms instead of 93 ms,
- it sends 6,562 bytes instead of 346 bytes,

- it needs about 66 KB more memory.

After connecting, both versions cost the same, and replacing the keys is faster with the post-quantum version (25 ms instead of 43 ms).

Source: our own measurements, *Post-Quantum VPN Benchmark Report*, sections "Results at a glance" and "Full results", in the project repository at `PQC_VPN/benchmarking/system_benchmark/benchmark_report.pdf`.

A meter connects once and then stays connected, so a cost that is paid only at connection time suits this use well.

9 Limits of this study

- We found no source saying that post-quantum meters are installed in homes today. The industry is at the stage of standards, chips and trials.
- The news about Suite 3 comes from a company that sells chips for it. We could not read the DLMS User Association's own paper.
- The sources do not say which sizes of ML-KEM and ML-DSA the meters will use. We used the smallest ones (ML-KEM-512 and ML-DSA-44).
- The three steps of a Suite 1 meter are taken from one research paper, which built a test meter. In that test the keys were placed in the code by hand.
- Real meters use a small ARM chip with an extra secure chip. We used an ESP32 without one.
- The NIST dates come from a draft report.

Sources

1. A. A. Mujtaba, B. Khowaja and S. Z. H. Rizvi, *Lightweight Cryptography Based VPN Framework for Secure IoT Communication*, capstone thesis, Habib University, Karachi, May 2026. Pages 12 and 15.

2. T. Huang, C. Chien, C. Wang and I. Liao, "The Implementation of an HSM-Based Smart Meter for Supporting DLMS/COSEM Security Suite 1", *Proc. 8th Int. Conf. on Internet of Things, Big Data and Security (IoTBDS 2023)*, pages 123–130. Pages 123, 124, 126, 127 and 128. https://www.scitepress.org/Papers/2023/118253/118253.pdf

3. NIST, *Transition to Post-Quantum Cryptography Standards*, NIST IR 8547 (initial public draft), November 2024. Pages 1, 13 and 14. https://nvlpubs.nist.gov/nistpubs/ir/2024/NIST.IR.8547.ipd.pdf

4. Landis+Gyr, "The Quantum Risk: Why Utilities Must Act Now to Secure the Grid", 20 October 2025. https://www.landisgyr.com/us/en/home/knowledge/blog/2025/The-Quantum-Risk--Why-Utilities-Must-Act-Now-to-Secure-the-Grid

5. SEALSQ, "Why Smart Metering Security Must Evolve Beyond PQC". https://www.sealsq.com/sealsq-blog/why-smart-metering-security-must-evolve-beyond-pqc

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7. PQShield, “ML-KEM and ML-DSA for ARM Cortex M CPU in smart meters”. https://pqshield.com/?p=9909

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9. Husarnet, “How Husarnet works”. https://husarnet.com/docs/how-husarnet-works/ and source code, https://github.com/husarnet/husarnet

10. Cloudflare, “Post-quantum IPsec interoperability with third-party devices”, changelog entry of 30 April 2026. https://developers.cloudflare.com/changelog/post/2026-04-30-ipsec-post-quantum-third-party/index.md